Year 10 Mathematics

Number 2

4.2 Recurring Decimals

Recurring Decimals

STARTER · 5 MIN

Date: _____ **Title:** Recurring Decimals

LEARNING OBJECTIVES

- Write fractions as recurring decimals
- Use dot notation for recurring decimals
- Convert recurring decimals to fractions

STARTER *Answer in your book. No calculators.*

1. Write as a decimal: $\frac{1}{4}$
2. Write as a decimal: $\frac{3}{8}$
3. Simplify $\frac{18}{99}$
4. Work out 100×0.36

Starter — Answers

STARTER

Tick or correct in green pen

1. $1 \div 4 =$ **0.25**

2. $3 \div 8 =$ **0.375**

3. $\frac{18}{99} =$ $\frac{\mathbf{2}}{\mathbf{11}}$

4. $100 \times 0.36 =$ **36**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can write fractions as recurring decimals	☐
SILVER	I can use dot notation for recurring decimals	☐
GOLD	I can convert recurring decimals to fractions	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Recurring Decimals

INTRODUCTION · 3 MIN

A recurring decimal has digits that repeat for ever.

Terminating

The decimal stops.

$\frac{3}{8} = 0.375$

Recurring

The digits repeat for ever.

$\frac{1}{3} = 0.333\ldots$

Dot notation

Put a dot over the first and last repeating digits.

$0.272727\ldots = 0.\dot{2}\dot{7}$

To a fraction

Let x = the decimal. Multiply by 10 or 100, then subtract.

$x = 0.\dot{7}, \quad 10x = 7.\dot{7}$

Examples 1 and 2 — Fractions to Recurring Decimals

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Write as a recurring decimal:

$$\frac{1}{3}$$

Example 2

Write as a recurring decimal:

$$\frac{5}{11}$$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$1 \div 3 = 0.333\ldots$

Answer : $\mathbf{0.\dot{3}}$

Tip: divide the top by the bottom.

Example 2

$5 \div 11 = 0.454545\ldots$

Answer : $\mathbf{0.\dot{4}\dot{5}}$

Tip: dot the first and last digit of the repeating block.

Task 1

PRACTICE · 9 MIN

BRONZE

Write each fraction as a recurring decimal using dot notation

1. $\frac{2}{3}$

2. $\frac{1}{9}$

3. $\frac{7}{11}$

4. $\frac{1}{6}$

5. $\frac{4}{7}$

★ Write 0.272727… using dot notation

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. $2 \div 3 = 0.666\ldots = \mathbf{0.\dot{6}}$
2. $1 \div 9 = 0.111\ldots = \mathbf{0.\dot{1}}$
3. $7 \div 11 = 0.6363\ldots = \mathbf{0.\dot{6}\dot{3}}$
4. $1 \div 6 = 0.1666\ldots = \mathbf{0.1\dot{6}}$
5. $4 \div 7 = 0.571428\ldots = \mathbf{0.\dot{5}71428\dot{}}$

★ $0.2727\ldots = \mathbf{0.\dot{2}\dot{7}}$

Examples 3 and 4 — Recurring Decimals to Fractions

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Write as a fraction:

$0.\dot{7}$

Example 4

Write as a fraction in its simplest form:

$0.\dot{3}\dot{6}$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x = 0.777\ldots$$
$$10x = 7.777\ldots$$
$$9x = 7$$
$$\mathbf{x = \frac{7}{9}}$$

Tip: one repeating digit: multiply by 10.

Example 4

$$x = 0.3636\ldots$$
$$100x = 36.3636\ldots$$
$$99x = 36$$
$$x = \frac{36}{99}$$
$$\mathbf{= \frac{4}{11}}$$

Tip: two repeating digits: multiply by 100.

Task 2

PRACTICE · 9 MIN

SILVER

Write each recurring decimal as a fraction in its simplest form

1. $0.\dot{5}$

2. $0.\dot{8}$

3. $0.\dot{2}\dot{7}$

4. $0.1\dot{8}$

5. $0.\dot{4}\dot{5}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $9x = 5, \quad x = \mathbf{\frac{5}{9}}$

2 $9x = 8, \quad x = \mathbf{\frac{8}{9}}$

3 $99x = 27, \quad x = \frac{27}{99} = \mathbf{\frac{3}{11}}$

4 $99x = 18, \quad x = \frac{18}{99} = \mathbf{\frac{2}{11}}$

5 $99x = 45, \quad x = \frac{45}{99} = \mathbf{\frac{5}{11}}$

Examples 5 and 6 — Harder Recurring Decimals

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5

(3 marks)

Write as a fraction in its simplest form:

$0.1\dot{6}$

Example 6

(2 marks)

Prove that $0.\dot{2}\dot{4} = \frac{8}{33}$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$10x = 1.666\ldots$$
$$100x = 16.666\ldots$$
$$90x = 15$$
$$x = \frac{15}{90}$$
$$= \mathbf{\frac{1}{6}}$$

Tip: make two numbers with the same repeating part, then subtract.

Example 6

$$x = 0.2424\ldots$$
$$100x = 24.2424\ldots$$
$$99x = 24$$
$$x = \frac{24}{99}$$
$$= \mathbf{\frac{8}{33}}$$

Tip: 'prove' means show every step.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Write as a fraction: $0.2\dot{7}$ (3 marks)

2. Write as a fraction: $0.3\dot{6}$ (3 marks)

3. Prove that $0.\dot{1}\dot{5} = \frac{5}{33}$ (2 marks)

4. Write as a fraction: $0.\dot{1}2\dot{3}$ (3 marks)

5. Prove that $0.4\dot{5} = \frac{41}{90}$ (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $100x - 10x = 27.7\ldots - 2.7\ldots, \quad 90x = 25, \quad x = \frac{25}{90} = \mathbf{\frac{5}{18}}$

2 $100x - 10x = 36.6\ldots - 3.6\ldots, \quad 90x = 33, \quad x = \frac{33}{90} = \mathbf{\frac{11}{30}}$

3 $100x - x = 15, \quad 99x = 15, \quad x = \frac{15}{99} = \mathbf{\frac{5}{33}}$

4 $1000x - x = 123, \quad 999x = 123, \quad x = \frac{123}{999} = \mathbf{\frac{41}{333}}$

5 $100x - 10x = 45.5\ldots - 4.5\ldots, \quad 90x = 41, \quad x = \mathbf{\frac{41}{90}}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write as a recurring decimal: $\frac{2}{9}$

SILVER Write as a fraction: $0.\dot{3}\dot{9}$

GOLD Write as a fraction: $0.1\dot{2}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $2 \div 9 = 0.222\ldots = \mathbf{0.\dot{2}}$

SILVER $99x = 39, \quad x = \frac{39}{99} = \mathbf{\frac{13}{33}}$

GOLD $90x = 11, \quad x = \mathbf{\frac{11}{90}}$

Exam Practice — Fractions to Decimals

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

Write each fraction as a recurring decimal.

(a) Use dot notation. (1)

$\frac{2}{3}$

.....

(b) (1)

$\frac{5}{11}$

.....

Question 2

(2 marks)

Write these numbers in order of size.
Start with the smallest number.

$0.35 \quad 0.\dot{3}\dot{5} \quad 0.\dot{3} \quad 0.355$

.....

Exam Practice — Recurring Decimals to Fractions

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

Write this recurring decimal as a fraction.

$$0.\dot{7}$$

.....

Question 4

(3 marks)

Show that the recurring decimal below can be written as a fraction in its simplest form.

You must show all your working.

$$0.\dot{2}\dot{7} = \frac{3}{11}$$

Exam Practice — Recurring Decimal Proofs

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

Prove algebraically that

$$0.1\dot{5} = \frac{7}{45}$$

Question 6

(3 marks)

Prove algebraically that

$$0.0\dot{3}\dot{6} = \frac{2}{55}$$

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $0.\dot{6}$ — B1

(b) $0.\dot{4}\dot{5}$ — B1

Tip: dot the first and last repeating digits.

Question 2

0.333…, 0.35, 0.3535…, 0.355 — M1

$0.\dot{3}$, 0.35, $0.\dot{3}\dot{5}$, 0.355 — A1

Tip: write each number to 4 d.p. first.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$x = 0.777\ldots,\ 10x = 7.777\ldots$ **M1**

$9x = 7$

$\mathbf{x = \frac{7}{9}}$ **A1**

Tip: one recurring digit → multiply by 10.

Question 4

$x = 0.2727\ldots,\ 100x = 27.27\ldots$ **M1**

$99x = 27$ **M1**

$\mathbf{x = \frac{27}{99} = \frac{3}{11}}$ **A1**

Tip: two recurring digits → multiply by 100.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x = 0.1555\ldots$

$100x = 15.55\ldots,\ \ 10x = 1.55\ldots$ — M1

$90x = 14$ — M1

$\mathbf{x = \frac{14}{90} = \frac{7}{45}}$ — A1

Tip: line up the recurring parts, then subtract.

Question 6

$x = 0.03636\ldots$

$1000x = 36.36\ldots,\ \ 10x = 0.36\ldots$ — M1

$990x = 36$ — M1

$\mathbf{x = \frac{36}{990} = \frac{2}{55}}$ — A1

Tip: subtract two multiples with the same decimals.